Artificial Intelligence - Lab 12 Solution

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Lab Task:
Develop Model’s Application Phase in Flask and use frontend code for user interaction. This solution shows how to connect a Machine Learning model with a Flask web application.

Project Folder Structure

```
project/
■■■ app.py
■■■ model.pkl
■■■ templates/
■ ■■■ index.html
■■■ static/
■■■ style.css
```

1. app.py

```
from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

# Load trained model
model = pickle.load(open('model.pkl', 'rb'))

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/predict', methods=['POST'])
def predict():
    feature1 = float(request.form['feature1'])
    feature2 = float(request.form['feature2'])

    features = np.array([[feature1, feature2]])

    prediction = model.predict(features)

    return render_template('index.html',
                           prediction_text=f"Prediction Result: {prediction[0]}")

if __name__ == '__main__':
    app.run(debug=True)
```

2. templates/index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Prediction System</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>

<div class="container">
    <h1>Machine Learning Prediction System</h1>

    <form action="/predict" method="POST">
        <input type="text" name="feature1" placeholder="Enter Feature 1" required>
```

```
        <input type="text" name="feature2" placeholder="Enter Feature 2" required>

        <button type="submit">Predict</button>
    </form>

    <h2>{{ prediction_text }}</h2>
</div>

</body>
</html>
```

3. static/style.css

```
body{
    font-family: Arial;
    background: #f2f2f2;
    text-align: center;
}

.container{
    width: 400px;
    margin: auto;
    margin-top: 100px;
    background: white;
    padding: 20px;
    border-radius: 10px;
}

input{
    width: 90%;
    padding: 10px;
    margin: 10px;
}

button{
    padding: 10px 20px;
    background: blue;
    color: white;
    border: none;
}
```

Steps to Run Flask Application

Step	Description
1	Install Flask using: pip install flask
2	Place all files in the project folder
3	Run command: python app.py
4	Open browser and visit: http://127.0.0.1:5000

Conclusion:
In this lab, a Flask web application was developed to connect a Machine Learning model with a frontend interface. The application takes user input, sends it to the trained model, and displays prediction results on the webpage.